The Reasoning Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: MEASURE, ANGLES, LOGIC and PROOF (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (MEASURE)
- ☐ **A2** (ANGLES)
- ☐ **A3** (LOGIC)
- ☐ **A4** (PROOF)
- ☐ **A5** (MEASURE)
- ☐ **A6** (LOGIC)

Entrées

10 pts each · choose 4

- ☐ **E1** (MEASURE)
- ☐ **E2** (MEASURE)
- ☐ **E3** (ANGLES)
- ☐ **E4** (ANGLES)
- ☐ **E5** (LOGIC)
- ☐ **E6** (PROOF)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (MEASURE)
- ☐ **S2** (ANGLES)
- ☐ **S3** (LOGIC)
- ☐ **S4** (PROOF)

My skill check (circle each skill you have covered): MEASURE · ANGLES · LOGIC · PROOF

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 MEASURE ☐ My pick · 5 pts

Find the midpoint of the segment with endpoints A(−2, 5) and B(6, 1).

A2 ANGLES ☐ My pick · 5 pts

Two angles form a linear pair. One measures (x + 20)° and the other measures 100°. Find x.

A3 LOGIC ☐ My pick · 5 pts

Write the **converse** of: “If a number is divisible by 6, then it is even.” Is the converse true or false?

A4 PROOF ☐ My pick · 5 pts

Give a counterexample to the statement: “If $x^2 > 9$, then $x > 3$.”

A5 MEASURE ☐ My pick · 5 pts

Find the distance between P(1, 2) and Q(4, 6).

A6 LOGIC ☐ My pick · 5 pts

Write the **contrapositive** of: “If a figure is a square, then it is a rectangle.”

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 MEASURE ☐ My pick · 10 pts

M(1, −2) is the midpoint of segment AB, and A is (−3, 4). Find the coordinates of B and show how you know.

E2 MEASURE ☐ My pick · 10 pts

Points A, B, C are collinear with B between A and C. AB = 2x + 3, BC = 4x − 5, and AC = 40. Find x, AB, and BC. Is B the midpoint of AC? Explain.

E3 ANGLES ☐ My pick · 10 pts

Two lines intersect. One pair of vertical angles measures (4x − 7)° and (2x + 19)°. Find x, the measure of these angles, and the measure of an angle adjacent to one of them.

E4 ANGLES ☐ My pick · 10 pts

Ray BD bisects ∠ABC. m∠ABD = (3x + 5)° and m∠DBC = (5x − 11)°. Find x and m∠ABC.

E5 LOGIC ☐ My pick · 10 pts

Statement: “If a figure is a rhombus, then it is a parallelogram.” Write its converse, inverse, and contrapositive. Decide whether each of the four is true or false, and give a counterexample for each false one.

E6 PROOF ☐ My pick · 10 pts

Given: ∠1 and ∠2 are supplementary, and ∠2 and ∠3 are supplementary. Prove m∠1 = m∠3. Give a reason for each step.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 MEASURE ☐ My pick · 20 pts

Segment AB has midpoint M(2, 1) and endpoint A(−2, −3). **(a)** Find B. **(b)** Find AB exactly as a simplified radical. **(c)** Let C = (6, −3). Show that triangle ABC is isosceles.

S2 ANGLES ☐ My pick · 20 pts

Lines AC and BD intersect at O. m∠AOB = (4x + 10)° and m∠BOC = (2x + 20)°. **(a)** Find x. **(b)** Find all four angles at O and name the reason for each. **(c)** Ray OE bisects ∠BOC. Find m∠AOE.

S3 LOGIC ☐ My pick · 20 pts

True statements: “If a team wins, then it advances.” “If a team advances, then it plays in the finals.” **(a)** Combine them into one if-then statement and name the law. **(b)** The Tigers did not play in the finals. What follows? **(c)** Rosa says, “The Lions played in the finals, so they won.” Explain her error.

S4 PROOF ☐ My pick · 20 pts

Given: A, B, C are collinear with B between A and C. M is the midpoint of AB and N is the midpoint of BC. Prove: MN = $\frac{1}{2}$AC. Write a proof with a reason for every step.